State ECEC demand targets: method and sources

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1 Purpose

This build estimates annual hours of early childhood education and care (ECEC) in six categories. It covers Kentucky (KY), New Jersey (NJ), New York State (NYS), New York City (NYC), and Virginia (VA). The method has six steps:

1. Estimate how many children receive regular care.
2. Divide them between formal and informal care.
3. Assign program counts to Unpaid Center-Based.
4. Divide the rest of formal care between centers and homes.
5. Divide informal care among its three types.
6. Convert each allocation to annual hours.

The build writes one six-row file per geography. Each row is modeled annual hours of care. The build does not assign individual children in a state.

In the NSECE survey, each child has one primary arrangement. Only the hours in that arrangement count. From those children, I estimate shares and mean hours by Census region. The build then divides a modeled state total of child-equivalents with those shares, and converts it to hours.

Annual hours by care type, 2019 (millions)

Care type	KY	NJ	NYS	NYC	VA
Center-Based	68.8	174.9	390.3	184.3	148.3
Unpaid Center-Based	70.9	79.5	143.7	64.0	77.6
Paid Home-Based	17.0	27.6	61.6	29.1	36.6
Other Unpaid	87.7	230.9	484.0	224.9	142.5
Other Paid	55.3	68.1	142.8	66.4	89.8
Unpaid Home-Based	6.3	11.2	23.5	10.9	10.2

The rows are defined by where the care happens and whether the family pays for it. Paid means the household reports a positive weekly payment for that arrangement. Individual care in the child's home, or by a caregiver with a prior relationship to the child, is Other.

The National Survey of Children's Health (NSCH) gives the share of children under five in regular non-parental care in each state. The National Survey of Early Care and Education (NSECE) records the type of care and divides those children among the six categories, by Census region.

The six care types, by setting and payment		
Setting	Family pays	Family does not pay
A center	Center-Based	Unpaid Center-Based
Individual care outside home; no prior relationship	Paid Home-Based	Unpaid Home-Based
Other individual/non-center care	Other Paid	Other Unpaid

2 The build

Step 1. Children in regular care

$$\text{control total} = \text{population under 5} \times \text{NSCH participation rate} \tag{1}$$

$$\text{NSCH participation rate} = \frac{\text{children under 5 whose answer is yes, with NSCH weights}}{\text{children under 5, with NSCH weights}} \tag{2}$$

- Population: Census population estimates, 2019, ages 0–4, summed from counties.
- Rate: NSCH 2019, question K6Q20: "Does the child receive care for at least 10 hours a week from someone other than a parent or guardian?" The question names every type of arrangement and does not ask which one applies. I use NSCH weights by state.
- NYC uses the NYS rate.

NSCH participation rate, children under 5, 2019				
	NSCH records	Rate	95% interval	Change in annual hours
KY	147	59.4%	49.6–69.2%	±17%
NJ	101	60.2%	47.5–72.9%	±22%
NYS	112	57.7%	46.1–69.2%	±20%
NYC	(NYS)	57.7%	46.1–69.2%	±20%
VA	121	52.0%	40.5–63.6%	±23%

NSCH records are unweighted children. The change in annual hours is the change in the build's total when the rate moves to its interval's extremes. I estimate the standard error with the NSCH survey design. NSCH samples addresses likely to hold a child at a higher rate than other addresses. The calculation keeps these two groups separate. For the hours interval, I run the build again at each end of the rate interval.

Step 2. Formal and informal totals

$$\text{formal} = \text{control total} \times \text{regional formal share} \qquad \text{informal} = \text{control total} - \text{formal} \tag{3}$$

$$\text{regional formal share} = \frac{\text{weighted children whose primary type is formal}}{\text{weighted children in regular care}} \tag{4}$$

Children in regular care are NSECE children aged 0–4 whose regular non-parental care totals at least 10 hours a week, with school excluded. An arrangement is regular when the household uses it on a regular schedule. NSECE screens for care starting at 5 hours. I restrict the analysis to children with at least 10 hours to match the NSCH threshold. Formal care is Center-Based, Unpaid Center-Based, and Paid Home-Based. Informal care is the other three types.

The formal share is 57.4% in the Northeast (NJ, NYS, NYC) and 54.1% in the South (KY, VA).

Every NSECE constant in this build — the formal share, the center-home split (step 4), the informal shares (step 5), and the weekly hours (step 6) — comes from the NSECE 2019 household file, with child weights by Census region. When a region has fewer than 30 NSECE records for a type, I use the national value.

Each child gets one primary type, in this order: Center-Based, Unpaid Center-Based, Paid Home-Based, Other Paid, Unpaid Home-Based, Other Unpaid. Between two paid centers, the higher-priced one wins. Hours decide all other ties. The simulator uses the same rule.

Step 3. Program care

I assign public pre-K and Head Start to Unpaid Center-Based:

$$\text{Unpaid Center-Based} \;=\; \text{pre-K} + \text{Head Start} + \text{Early Head Start} \tag{5}$$

$$\text{other formal} \;=\; \text{formal} \;-\; \text{Unpaid Center-Based} \tag{6}$$

Program counts, 2018–19

	Public pre-K (enrolled)	Head Start (slots)	Early Head Start (slots)	Total
KY	28,465	12,307	3,004	43,776
NJ	41,305	11,940	3,578	56,823
NYS	54,451	36,863	11,433	102,747
NYC	26,717	13,073	5,938	45,728
VA	33,790	11,579	2,507	47,876

- Public pre-K is pre-K grade enrollment from the Common Core of Data (CCD) of the National Center for Education Statistics (NCES). It counts a grade, not an age, so includes some five-year-olds.
- Head Start serves children from age 3 until they start kindergarten. Early Head Start serves children under 3. The slot counts do not give ages, so the National Institute for Early Education Research (NIEER) estimates them. NIEER counts five-year-olds in Head Start as four-year-olds, so the Head Start count includes some five-year-olds.
- NYC's Head Start count is the city's share of NYS funded slots in the Head Start site file of the Administration for Children and Families (ACF): 7,194 of 20,285. The site file is from 2026. I assume the city's share of slots was the same in 2019.

- I do not deduct overlap between Head Start and pre-K, despite the fact that some Head Start runs inside public schools. The school district can count those children in pre-K enrollment. The same child can appear in both counts.

Step 4. Centers and homes

$$\text{Center-Based} \;=\; \text{other formal} \times \text{regional center share} \tag{7}$$

$$\text{Paid Home-Based} \;=\; \text{other formal} \times (1 - \text{regional center share}) \tag{8}$$

$$\text{regional center share} \;=\; \frac{\text{weighted children in paid centers}}{\text{weighted children in paid centers or Paid Home-Based}} \tag{9}$$

The center share is 0.876 in the Northeast and 0.805 in the South.

Step 5. Informal care

$$\text{informal share of a type} \;=\; \frac{\text{weighted children of that primary type}}{\text{weighted children with an informal primary type}} \tag{10}$$

Informal care accounts for 43–46% of the estimated children in regular care, but 48–52% of modeled annual hours as its average weekly hours are higher.

Step 6. Annual hours

$$\text{annual hours} \;=\; \text{allocated child-equivalents} \times \text{average weekly hours} \times 50 \tag{11}$$

$$\text{average weekly hours of a type} \;=\; \frac{\text{weighted sum of primary-arrangement hours}}{\text{weighted children of that primary type}} \tag{12}$$

NSECE regional constants, 2019

	Northeast			South		
Care type	Share of informal	Weekly hours	NSECE records	Share of informal	Weekly hours	NSECE records
Center-Based	—	33.0	118	—	39.0	264
Unpaid Center-Based	—	28.0	84	—	32.4	163
Paid Home-Based	—	36.7	33	—	39.8	55
Other Unpaid	0.632	55.4	116	0.560	42.1	294
Other Paid	0.329	31.3	60	0.401	37.1	207
Unpaid Home-Based	0.039*	43.4*	8	0.039*	43.4*	4

- *National value: the region has fewer than 30 records. The other two informal shares are rescaled so the three sum to 1.
- 50 weeks matches the simulator's assumption of two weeks without care.
- Records are the unweighted NSECE children behind each row. Precision differs by row.

3 How to interpret the results

1. The results are modeled annual hours, not observed children.
2. The simulator can imply different child counts. It uses fixed part-time and full-time hours, and step 6 uses a mean for each type.
3. Geographies in the same region differ only through population, the NSCH rate, and the program counts.
4. Program counts stand for Unpaid Center-Based. No source records what a family pays for a program place.
5. The NYS file includes the NYC children. NYS and NYC use the same rate and Northeast values. The NYS rows minus the NYC rows give the non-NYC part of the state.
6. The targets include only care of at least 10 hours a week, but the simulator's baseline includes every regular arrangement. Children under 10 hours are 8.5% of the children in care but 1.5% of the hours.
7. The NSECE shares and hours carry no interval.

4 Open questions

1. *Pool NSCH waves?* The build uses the 2019 wave alone. Pooling with 2018, 2020, or both cuts the standard error:

Rate (standard error)	2019	2018–19	2019–20	2018–20
KY	59.4 (5.0)	58.3 (3.7)	53.1 (3.4)	54.5 (2.9)
NJ	60.2 (6.5)	59.2 (4.3)	50.7 (3.9)	53.4 (3.3)
NYS	57.7 (5.9)	61.2 (4.3)	59.9 (3.9)	61.5 (3.3)
VA	52.0 (5.9)	57.0 (4.1)	49.3 (3.9)	53.5 (3.2)

   NSCH 2020 was fielded through the pandemic. The 2018–19 pool avoids 2020, but centers on mid-2018, not the 2019 base year.

2. *Find enrolled slots that are in use?* An earlier build used each state's provider register to split formal care between centers and homes. However, it required fill rates, age adjustments, rebasing 2026 registers to 2019, and an address match to remove program children already in a register. The registers also covered different providers in each state. The build now uses the regional NSECE split. A state source of enrolled slots that are in use, by setting, would let the build return to a state-specific split without those assumptions.
3. *Mean or median weekly hours?* Step 6 uses the mean because annual hours is a total. The modeled count times the estimated mean gives an estimated total; the median does not target that total. Weekly hours bunch at full time, with a tail of part-time care, so the median would move the totals. The mean is sensitive to extreme values in small cells.
4. *Regional or national NSECE constants?* The build uses regional constants to reflect differences observed between regions. But some regional cells are small. National constants rest on more

records, but give every region the same mix. Section 5 compares the two, and a hybrid of regional shares with national hours.

5 Sensitivity: national NSECE constants

I run the same six steps two more ways, and compare each with the regional default. Population, the NSCH rate, and the program counts do not change. The national build makes every NSECE constant national: a formal share of 0.545, a center share of 0.791, and the national informal mix and weekly hours. The hybrid build keeps the regional shares and uses the national weekly hours.

Both builds cut total hours by 3.0 to 3.9%. Only the national build raises Northeast Paid Home-Based, by about 63%, because its center share is lower. Both builds cut Northeast Other Unpaid by about 20%, because its national weekly hours are lower (44.2 versus 55.4 in the Northeast). In every build, the NSCH rate's 95% interval moves total hours ±17% (KY) to ±23% (VA). This covers NSCH sampling error only. Change in the table is each build's hours over the regional hours, minus 1.

Annual hours, regional, hybrid, and national NSECE constants, 2019 (millions)

	Regional			Change	
Care type	(default)	Hybrid	National	Hybrid	National
Kentucky (KY)					
Center-Based	68.8	63.3	63.2	−8.1%	−8.2%
Unpaid Center-Based	70.9	65.8	65.8	−7.2%	−7.2%
Paid Home-Based	17.0	16.4	17.9	−3.5%	+5.0%
Other Unpaid	87.7	92.1	96.1	+5.0%	+9.5%
Other Paid	55.3	50.4	46.3	−8.8%	−16.4%
Unpaid Home-Based	6.3	6.3	6.2	0.0%	−1.0%
Total	306.1	294.3	295.4	−3.8%	−3.5%
New Jersey (NJ)					
Center-Based	174.9	190.2	159.3	+8.8%	−8.9%
Unpaid Center-Based	79.5	85.4	85.4	+7.5%	+7.5%
Paid Home-Based	27.6	28.9	45.0	+4.7%	+63.0%
Other Unpaid	230.9	184.5	183.9	−20.1%	−20.4%
Other Paid	68.1	73.6	88.5	+8.1%	+30.0%
Unpaid Home-Based	11.2	11.2	11.9	0.0%	+6.7%
Total	592.1	573.9	574.0	−3.1%	−3.1%
New York State (NYS)					
Center-Based	390.3	424.6	357.3	+8.8%	−8.5%
Unpaid Center-Based	143.7	154.4	154.4	+7.5%	+7.5%
Paid Home-Based	61.6	64.5	100.9	+4.7%	+63.7%
Other Unpaid	484.0	386.8	385.5	−20.1%	−20.4%
Other Paid	142.8	154.4	185.6	+8.1%	+30.0%
Unpaid Home-Based	23.5	23.5	25.0	0.0%	+6.7%

Annual hours, regional, hybrid, and national NSECE constants, 2019 (millions)					
	Regional			Change	
Care type	(default)	Hybrid	National	Hybrid	National
Total	1,245.9	1,208.2	1,208.7	−3.0%	−3.0%
New York City (NYC)					
Center-Based	184.3	200.5	168.9	+8.8%	−8.4%
Unpaid Center-Based	64.0	68.7	68.7	+7.5%	+7.5%
Paid Home-Based	29.1	30.5	47.7	+4.7%	+63.9%
Other Unpaid	224.9	179.8	179.1	−20.1%	−20.4%
Other Paid	66.4	71.7	86.3	+8.1%	+30.0%
Unpaid Home-Based	10.9	10.9	11.6	0.0%	+6.7%
Total	579.5	562.1	562.3	−3.0%	−3.0%
Virginia (VA)					
Center-Based	148.3	136.3	135.7	−8.1%	−8.5%
Unpaid Center-Based	77.6	71.9	71.9	−7.2%	−7.2%
Paid Home-Based	36.6	35.4	38.3	−3.5%	+4.6%
Other Unpaid	142.5	149.6	156.0	+5.0%	+9.5%
Other Paid	89.8	81.9	75.1	−8.8%	−16.4%
Unpaid Home-Based	10.2	10.2	10.1	0.0%	−1.0%
Total	505.0	485.4	487.2	−3.9%	−3.5%

6 Sources

Sources of the inputs			
Input	Source	Data year	Population
Population 0–4	Census estimates, county	2019	All children 0–4
Participation rate	NSCH K6Q20	2019	Children 0–4 (NYC uses NYS)
Formal share	NSECE 0009, by region	2019	Children 0–4, ≥10 hours
Center-home split	NSECE 0009, by region	2019	Paid formal children, ≥10 hours
Informal mix	NSECE 0009, by region	2019	Informal children, ≥10 hours
Weekly hours	NSECE 0009, by region	2019	Children 0–4, ≥10 hours
Public pre-K	NCES CCD	2018–19	Pre-K grade
Head Start	NIEER yearbook	2018–19	Funded slots
Early Head Start	NIEER report	2018–19	Funded slots
Head Start sites	ACF site file	2026	Funded slots by site; NYC share of NYS Head Start